ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

TECHNOLOGY

PRACTICE EXAM 1 · MARKING GUIDELINE

MARKS: 75

Marking guideline for the Academic Central Grade 7 Technology practice exam 1.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words and other correct examples. In calculations give method marks even if the final answer is wrong.

QUESTION 1

1 1.1 B ✓ 1.2 C ✓ 1.3 A ✓ 1.4 B ✓ 1.5 D ✓ 1.6 A ✓ 1.7 C ✓ 1.8 C ✓ 1.9 A ✓ 1.10 D ✓ [10]

QUESTION 2

2.1 (a) Frame ✓ (b) Solid ✓ (c) Shell ✓ (3)

2.2 (a) Tension ✓ (b) Compression ✓ (c) Torsion ✓ (d) Bending ✓ (4)

2.3.1 Gate B ✓ (1)

2.3.2 The diagonal bar divides the rectangle into triangles ✓. A triangle cannot change shape unless a side breaks, so the gate cannot sag or be pushed into a parallelogram shape ✓. (2)

2.4 Any TWO: folding (e.g. corrugating) ✓; using tubes ✓; adding ribs or gussets; using stronger materials; adding struts and ties (2)

2.5 A static load does not move or change ✓, e.g. the weight of the bridge itself ✓. A dynamic load moves or changes ✓, e.g. cars and people crossing the bridge, or wind ✓. (4)

2.6 Any TWO: strong enough to carry its load ✓; stable (does not fall over) ✓; rigid; safe; uses materials economically; looks good (aesthetics); fit for its purpose (2)

[18]

QUESTION 3

3.1 (a) Transparent ✓ (b) Strong / high tensile strength ✓ (c) Flexible / elastic / waterproof / grips the road ✓ (d) Soft / absorbs sweat / comfortable ✓ (4)

3.2 (a) Any TWO: protects the product ✓; keeps food fresh (preserves) ✓; gives information (ingredients, dates); advertises the product; makes it easy to carry and stack
(b) To reduce waste in landfills and pollution ✓, and to save raw materials and energy ✓ (4)

3.3 Any TWO methods ✓✓, e.g. drying, salting, freezing, canning, pickling, pasteurising, refrigeration. Explanation, e.g. drying removes water so bacteria and fungi cannot grow; freezing slows down bacteria ✓ (3)

3.4 Tin snips (accept metal shears) ✓ (1)

[12]

QUESTION 4

4.1 Lever A: first class ✓ Lever B: second class ✓ Lever C: third class ✓ (3)

4.2 $\text{MA} = 600 \div 200 \checkmark = 3 \checkmark$ (2)

4.3 (a) $\text{MA} = 1{,}2 \div 0{,}3$ ✓ $= 4$ ✓ (b) $\text{Effort} = 800 \div 4$ ✓ $= 200$ N ✓ (4)

4.4 (a) Reverse-motion linkage ✓ (b) Bell crank ✓ (2)

4.5 Any TWO: bottle opener ✓; tongs ✓; nutcracker; tin opener; scissors; garlic press (2)

4.6 Input: hand squeezes the brake lever ✓ → Process: the cable pulls the brake arms (lever action) ✓ → Output: brake pads press on the wheel rim and the bicycle slows down ✓ (3)

4.7 A long spanner has a longer effort arm ✓, so a smaller effort gives a bigger turning effect (moment) on the nut ✓. (2)

[18]

QUESTION 5

5.1.1 A cell ✓ B switch ✓ C light bulb ✓ (3)

5.1.2 No ✓. The switch is open, so the circuit is incomplete and no current can flow ✓. (2)

5.2 (a) Two pairs of long and short parallel lines joined together ✓ (b) A semicircle (dome) on two leads, or the box-shaped buzzer symbol ✓ (2)

5.3 (a) Conductor ✓ (b) Insulator ✓ (c) Conductor ✓ (d) Insulator ✓ (4)

5.4 (a) Wind insulated copper wire many times around an iron nail ✓ and connect the ends to a cell ✓. (5)
(b) Any TWO: more turns of wire ✓; more cells (more current) ✓; use a soft iron core
(c) Any ONE: crane in a scrapyard; electric bell; door lock; loudspeaker; electric motor ✓

5.5 Any ONE: never touch switches or plugs with wet hands; do not overload plugs; do not use damaged cords; never push objects into plug sockets ✓ (1)

[17]

TOTAL: 75

END OF MARKING GUIDELINE